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Enterprise Case Studies: The Business Impact of Handhelds

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3Com
More connected.

Session Agenda

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- Introduction
- Case Studies
- Measuring Return
- Summary
- Q&A

Introduction

- Case Studies
  - Volvo Cars
  - United Center
  - USPS
- Business case study
  - Overview
  - Situation
  - Challenge
  - Solution
  - Results
- Attend Session 510

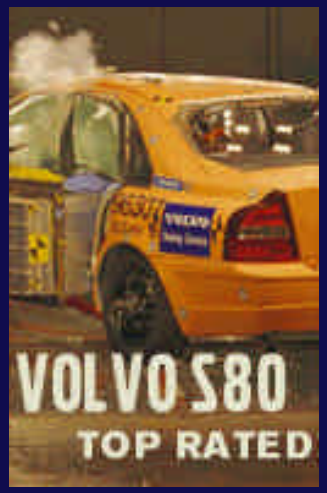
VOLVO

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Welcome to

VOLVO CAR CORPORATION









Safety, Quality, Care for the Environment

Volvo Cars

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The Organization

- Founded in 1927 in Sweden
- Owned today by Ford Motor Company
- 1.2M+ cars in N. America
- Revenue stream based on reputation for quality
- Quality assurance is rigorous
- Continuous inspections—manufacturing to showroom

"Cars are driven by people. The guiding principle behind everything we make at Volvo therefore is, and must remain, safety."

Assar Gabrielsson. A Volvo founder. 1927.

Volvo Cars

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The Situation–Port Operations Group

- 125,000 Volvos imported annually
- Arrive at four different U.S. ports
- Inspections documented on paper forms on metal clipboards
- Documentation
  - 17-digit vehicle ID
  - Date of inspection
- Inspection
  - 40 separate inspection items
  - List of items revised constantly

Volvo Cars

The Challenge

- Metal clipboards damage car finishes
- Potential for error with 17-digit VIN & data entry
- Constantly reprinting paper forms to keep current with latest inspection criteria
- Cumbersome and inefficient data entry process
- Slow access to data

What's at Stake?

Assuring Quality while Keeping Costs Down

Supporting a Mission-Critical Core Function

Volvo Paper-Based Process

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QA Inspectors Collect Data and Complete Inspections

Handsorted by Volvo Staff



To Vendor for Manual DB entry



Info Returned to Volvo

VOLVO

Process complete in 3 WEEKS!

Volvo Cars

The Solution

- Partner = TikiSoft, Inc.
- Device = Palm V™ organizer
- Software-Based Form
  - Check boxes and pick lists
- Reports can be generated daily, weekly, quarterly, YTD

TikiSoft
A SHARED VISION...

The Inspection system was designed with change in mind.

Management has the ability to easily make their own changes without our technical assistance or additional development costs.

Marc Weiss, TikiSoft

Volvo Palm Computing® Solution

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QA Inspectors Collect Data and Complete Inspections on Palm V™ Connected Organizers using Pick Lists and Check Boxes





Palm HotSync® Technology Used to Transfer Data to a Central Location

Data Available Immediately

Process complete in 1 DAY!

Volvo Cars

The Results

- Improved data collection
  - IMMEDIATE access to data—turnaround of 1 day
  - Increased data accuracy
- Eliminated costs
  - Updating and printing new paper forms
  - Manual data entry
  - Overnight mailings



Volvo Cars

The Bottom Line

The Palm Solution enables Volvo to fully leverage its quality control information to maintain its reputation as a provider of safe, high-quality automobiles





A tradition of delivering value

Delivering far and wide
Delivering the goods
Delivering customers
Delivering opportunities
Delivering hope



USPS

The Organization

- Delivers 650M+ pieces of mail every day
- Employs 775,000 workers
- 38,000 post offices, fleet of 200,000+ vehicles
- Use commercial airlines to carry 10.5M pounds of mail a day

41% of World's Mail Volume

Largest Civilian Employer

Biggest Airline Customer

USPS

The Situation

- Commercial airlines transport mail
- Mail bundled into receptacles
- Receptacles mishandled/damaged
- Ramp clerks create incident reports
- Data used to
  - Reduce mishandling of mail
  - Change airlines
  - Assess fines







USPS

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The Challenge

- Paper-based process performed days or weeks after the fact
- Prone to errors
- Information incomplete or illegible
- Delay of months before information provided to airlines for resolution

What's at Stake?

Ability to…

- **Resolve Vendor Quality Issues**
- **Gain Leverage with Vendors**
- **Manage Expenses in the Logistics Chain**
- **Compete by Delivering Value**

USPS Paper-Based Process



Mail receptacles damaged en route

USPS ramp clerks create incident reports

Incident reports delivered to central USPS location

Data consolidated

Results delivered to airline

Action taken
Problem resolved
Fees assessed
Vendor terminated

Process completed in MONTHS!

USPS

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The Solution

- Partner = CIBER
- Device = Symbol 1730 bar code scanner
- Connectivity = Riverbed ScoutMTS 3.0
- Barcode scanning, pick-lists, check-boxes
- Centralized USPS database
- Access over USPS Intranet



USPS Palm Computing® Solution

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Mail receptacles damaged en route

USPS ramp clerks scans mail receptacle



Ramp clerk uses pick-lists and check boxes to collect relevant data



At end of day, ramp clerk place devices in an Ethernet cradle and upload data

Results available immediately

Action taken
Problem resolved
Fees assessed
Vendor terminated

Process completed in DAYS!

USPS

The Results

- Increased accuracy and speed of access
- Enhanced decision making
- Improved quality
- Stronger vendor control
- Reduced data entry expenses



USPS

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The Bottom Line

The Palm™ Solution enables the USPS to continue its tradition of delivering value by improving its ability to manage commercial airline vendors



UNITED CENTER



CHICAGO BULLS

Sports & Entertainment Center

United Center

The Organization

- Massive sports and entertainment center
  - Home of the Chicago Bulls basketball team
  - Home of Chicago Blackhawks hockey team
- $175M in construction costs
- Opened in 1994
- Over 13M guests in five years
- Houses Fandemonium official team retail store



United Center

The Situation

- Fandemonium retail stores
- 50,000–300,000 items sold at a single event
- 2 retail outlets
- 12 merchandise stands throughout stadium



United Center

The Challenge

- No cash registers
- No way to collect sales information
  - Colors
  - Sizes
  - Styles
- No way to track
  - Lost or missing inventory
  - Distribution to each stand
- No efficient way to manage inventory



What's at Stake?

Lost Revenue Opportunities

Excessive Inventory Costs Affect Profits

Reduced Customer Satisfaction

United Center Process

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Fans rush to purchase 50,000–300,000 specialty items every event

Fandemonium merchandise stands accept cash and place in a cash box

Gross sales revenues reported at end of day

Inventory and accounting reconciliation takes place

Challenge
Out of stock of high-dema items—lost revenues

Challenge
Missing or lost inventory not accounted for

Challenge
No ability to requirements sales detail

Challenge
Expensive and time-consuming reconciliation process

Process complete in 6 DAYS!

United Center

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The Solution

- Partner = Braxton Butterfield Consulting, Inc.
- Device = Symbol 1700
- Database = Oracle8i Lite database
- Connectivity = Palm HotSync® technology
- Used in
  - Merchandise stands for POS data collection
  - Receiving dock for purchase order and inventory management

United Center
Palm Computing® Solution

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Fans rush to purchase 50,000–300,000 specialty items every event

Fandemonium merchandise use Symbol device to scan items sold and amounts collected



Data uploaded into Oracle8i Lite database

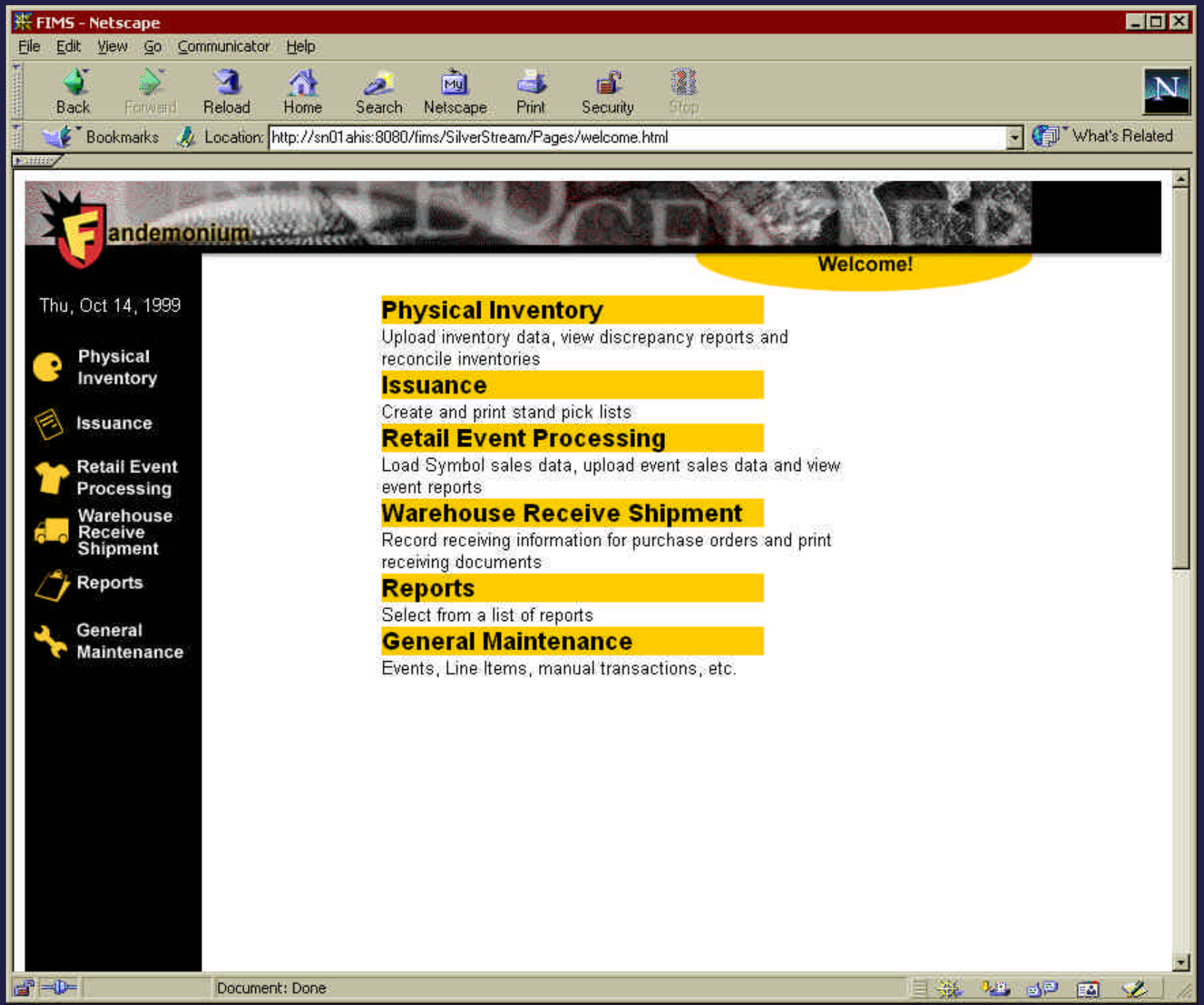
Oracle8*i*



Data imported into an Oracle8i database using HotSync® technology



Process complete in 30 MINUTES!


FIMS - Netscape
File Edit View Go Communicator Help
Back Forward Reload Home Search Netscape Print Security Stop
Bookmarks Location: http://sn01ahis:8080/fims/SilverStream/Pages/welcome.html
What's Related
Fandemonium
Welcome!
Thu, Oct 14, 1999
Physical Inventory
Issuance
Retail Event Processing
Warehouse Receive Shipment
Reports
General Maintenance
Physical Inventory
Upload inventory data, view discrepancy reports and reconcile inventories
Issuance
Create and print stand pick lists
Retail Event Processing
Load Symbol sales data, upload event sales data and view event reports
Warehouse Receive Shipment
Record receiving information for purchase orders and print receiving documents
Reports
Select from a list of reports
General Maintenance
Events, Line Items, manual transactions, etc.
Document: Done

United Center

The Results

- Efficient inventory management
  - Reduces on-hand inventory
  - Reduces time to reconcile inventory
  - Eliminates manual data entry
- Increased revenue
  - Meet customer demands by replenishing best-selling items



United Center

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The Bottom Line

The Palm™ Solution enables the United Center to increase its efficiency resulting in increased revenues and a better experience for the fans



Overview of Strategic Benefits

- Upholds key brand element
- Strengthens critical partnership
- Faster access means competitive advantage
- Provides data where none existed previously
- Streamlines a basic function

Ongoing commitment to the organizational identity is critical

Measuring Return

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ROI—Prevailing Perception

- Return on investment analysis
  - A simple tool to make sound business decisions
  - ROI a watchdog for traditional IT investments
- IT execs pressured to determine ROI
- Confidence levels
  - Majority believe ROI can work for IT
  - Remainder skeptical, but use it anyway
- Use of ROI varies by industry
  - Finance/banking vs. high tech

A strong ROI can makes things happen!

ROI is not an absolute justification…

it's a tool to select the most efficient solution

Source: Cambridge Information Networks, "ROI on IT Investments: Practical or Political?", 1999.

Measuring Return

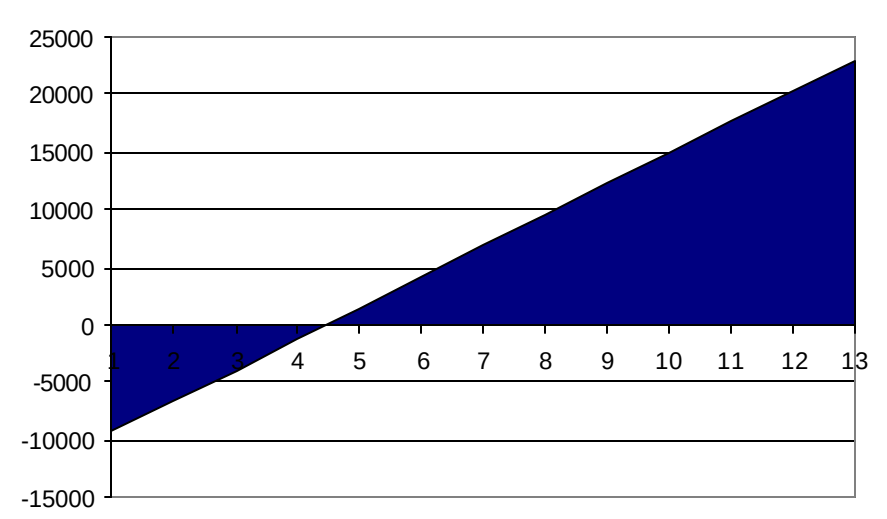
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A Sample Payback Analysis

Total Costs

/ Monthly Cash Flow

= Months until payback



Measuring Return

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A Simple ROI Calculation

Benefits

\- Costs

= Cash flow

/ Total Costs*

= ROI %

The ROI

- **An annual percentage that spans 3–4 years**
- **Look for a positive ROI!**

Year 1	Year 2	Year 3	Year 4
109%	159%	176%	184%

* **A strong ROI will also consider depreciation, taxes, and the time value of money**

Measuring Return

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Assessing Benefits

- Hard dollar benefits
  - Reduction in vendor mgmt. costs
  - Printing costs, mailing costs
  - Labor for manual data entry
- Soft benefits
  - Inventory mismanagement
  - Bad data
  - Customer perception/satisfaction
- Revenue increases
  - New sales opportunities
  - Higher margins

- Directly related to project
- Easily quantifiable
- Easy access to data

- Difficult access
- Delayed access
- Indirectly related to project

- Easily quantifiable
- May not be related at all!!

Measuring Return

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An Important Point....

Handhelds DO provide benefits in ALL these categories

As a developer working to get approval for your project...

you SHOULD position yourself to take credit anywhere you can!

Need Help with ROI?

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Bring us a success story,

we'll help you
calculate the ROI

success_story@palm.com

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Summary & Q&A

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